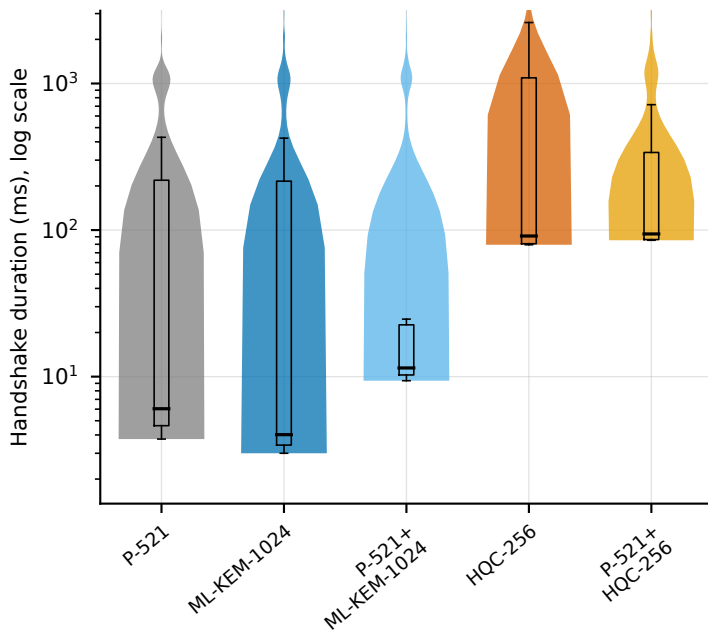
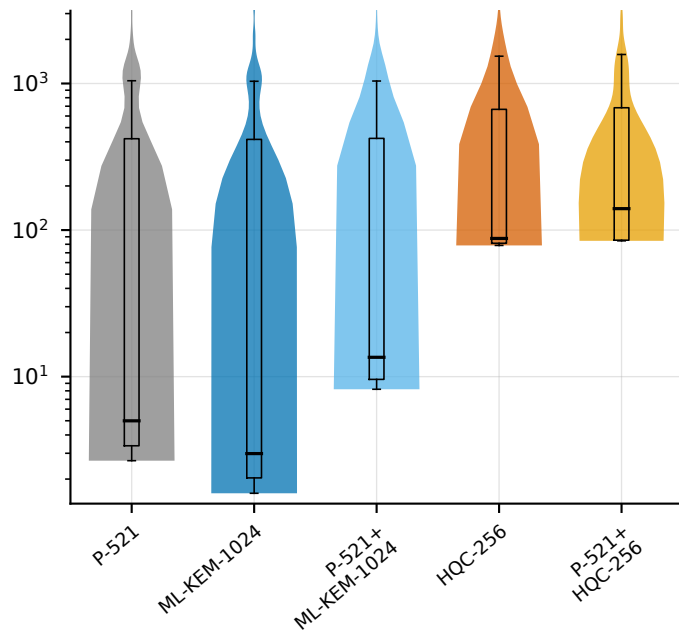


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	294.9	212.7	6.61k	406.9	217.7	13.44k
ML-KEM-1024	284.0	227.6	7.21k	378.6	194.9	7.42k
P-521+ML-KEM-1024	255.3	210.8	4.15k	444.0	320.3	26.43k
HQC-256	709.1	371.1	52.80k	594.9	287.4	30.47k
P-521+HQC-256	446.6	160.1	7.25k	577.8	171.6	6.88k